

Chapter 8:

Integration Techniques and Improper Integrals

Section 8.3: Trigonometric Integrals

Integrals of Powers of Sine and Cosine

We begin by considering integrals in which the integrand is a power of sine, a power of cosine, or a product of these.

$$\int \sin^m x \cos^n x \, dx$$

EXAMPLE The odd power is on cosine

Find $\int \cos^5 x \ dx$.

SOLUTION

$$I = \frac{1}{5}\sin^5 x - \frac{2}{3}\sin^3 x + \sin x + C$$

EXAMPLE **The odd power is on sine**

Find $\int \sin^3 x \sqrt{\cos x}\ dx.$

$$I = \frac{2}{7}\cos^{\frac{7}{2}} x - \frac{2}{3}\cos^{\frac{3}{2}} x + C$$

EXAMPLE **The powers on cosine and sine are even.**

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

Find $\int \sin^2 4x \ \ dx.$

SOLUTION

$$I = \frac{1}{2}x - \frac{1}{16}\sin(8x) + C$$

EXAMPLE

Find $\int_0^{\frac{\pi}{4}} \sqrt{1+\cos(4x)}\ dx.$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

SOLUTION

$$I = \frac{\sqrt{2}}{2}$$

EXAMPLE **The powers on cosine and sine are even.**

Find $\int \sin^2 x \cos^2 x \, dx$.

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

SOLUTION

$$I = \frac{1}{8}x - \frac{1}{32}\sin(4x) + C$$

GUIDELINES FOR EVALUATING INTEGRALS INVOLVING POWERS OF SINE AND COSINE

1. When the power of the sine is odd and positive, save one sine factor and convert the remaining factors to cosines. Then expand and integrate.

$$\int \sin^{\overbrace{2k+1}^{\text{Odd}}} x \cos^n x \, dx = \int (\overbrace{\sin^2 x)^k}^{\text{Convert to cosines}} \cos^n x \overbrace{\sin x \, dx}^{\text{Save for } du} = \int (1 - \cos^2 x)^k \cos^n x \sin x \, dx$$

2. When the power of the cosine is odd and positive, save one cosine factor and convert the remaining factors to sines. Then expand and integrate.

$$\int \sin^m x \cos^{\overbrace{2k+1}^{\text{Odd}}} x \, dx = \int (\sin^m x)(\overbrace{\cos^2 x)^k}^{\text{Convert to sines}} \overbrace{\cos x \, dx}^{\text{Save for } du} = \int (\sin^m x)(1 - \sin^2 x)^k \cos x \, dx$$

3. When the powers of both the sine and cosine are even and nonnegative, make repeated use of the formulas

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \text{and} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

to convert the integrand to odd powers of the cosine. Then proceed as in the second guideline.

EXAMPLE

Find $\int \sinh^3 x \ \cosh^2 x \ dx$.

SOLUTION

$$I = \frac{1}{5}\cosh^5 x - \frac{1}{3}\cosh^3 x + C$$

Integrals of Powers of Secant and Tangent

We use similar reasoning to evaluate integrals of the form

$$\int \tan^m x \sec^n x \, dx$$

We know that

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

EXAMPLE **The power on secant is even.**

Try to find $\int \cot^6 x \csc^4 x \, dx.$

Find $\int \tan^6 x \sec^4 x \, dx.$

SOLUTION

$$I = \frac{1}{9}\tan^9 x + \frac{1}{7}\tan^7 x + C$$

EXAMPLE **The power on tangent is odd.**

Try to find $\int \cot^3 x \csc^5 x \, dx.$

Find $\int \tan^5 x \sec^7 x \, dx.$

SOLUTION

$$I = \frac{1}{11}\sec^{11} x - \frac{2}{9}\sec^{9} x + \frac{1}{7}\sec^{7} x + C$$

EXAMPLE **The power on tangent is odd.**

Try to find $\int \cot^3 x \ \csc^{\frac{1}{2}} x \ dx.$

Find $\int \frac{\tan^5 x}{\sqrt{\sec x}} \, dx.$

SOLUTION

$$I = \frac{2}{7}\sec^{\frac{7}{2}} x - \frac{4}{3}\sec^{\frac{3}{2}} x - 2\sec^{-\frac{1}{2}} x + C$$

GUIDELINES FOR EVALUATING INTEGRALS INVOLVING POWERS OF SECANT AND TANGENT

1. When the power of the secant is even and positive, save a secant-squared factor and convert the remaining factors to tangents. Then expand and integrate.

$$\int \overbrace{\sec^{2k} x}^{\text{Even}} \tan^n x\, dx = \int \overbrace{(\sec^2 x)^{k-1}}^{\text{Convert to tangents}} \tan^n x \overbrace{\sec^2 x\, dx}^{\text{Save for } du} = \int (1 + \tan^2 x)^{k-1} \tan^n x \sec^2 x\, dx$$

2. When the power of the tangent is odd and positive, save a secant-tangent factor and convert the remaining factors to secants. Then expand and integrate.

$$\int \sec^m x \overbrace{\tan^{2k+1} x}^{\text{Odd}}\, dx = \int (\sec^{m-1} x) \overbrace{(\tan^2 x)^k}^{\text{Convert to secants}} \overbrace{\sec x \tan x\, dx}^{\text{Save for } du} = \int (\sec^{m-1} x)(\sec^2 x - 1)^k \sec x \tan x\, dx$$

3. When there are no secant factors and the power of the tangent is even and positive, convert a tangent-squared factor to a secant-squared factor, then expand and repeat if necessary.

$$\int \tan^n x\, dx = \int (\tan^{n-2} x) \overbrace{(\tan^2 x)}^{\text{Convert to secants}}\, dx = \int (\tan^{n-2} x)(\sec^2 x - 1)\, dx$$

4. When the integral is of the form

$$\int \sec^m x\, dx$$

where m is odd and positive, use integration by parts, as illustrated in Example 5 in Section 8.2.

5. When the first four guidelines do not apply, try converting to sines and cosines.

EXAMPLE **The power on tangent/cotangent is even.**

Find $\int \tan^4 x \, dx$.

SOLUTION

$$I = \frac{1}{3}\tan^3 x - \tan x + x + C$$

EXAMPLE Converting to Sines and Cosines

Find $\int \frac{\sec x}{\tan^2 x}\, dx.$

SOLUTION

$$I = -\csc x + C$$

EXAMPLE **The power on secant/cosecant is odd. (Here we need integration by parts)**

Find $\int \sec^3 x \; dx.$

SOLUTION **See slide # 13 in section 8.2**

$$I = \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$$

EXAMPLE **The power on secant/cosecant is odd and on tan is even. (Here we need integration by parts)**

Find $\int \tan^2 x \sec x \, dx.$

SOLUTION

$$I = \frac{1}{2}\sec x \tan x - \frac{1}{2}\ln|\sec x + \tan x| + C$$

Integrals Involving Sine-Cosine Products

The following product identities are useful in evaluating certain trigonometric integrals.

2 To evaluate the integrals (a) $\int \sin mx \cos nx\, dx$, (b) $\int \sin mx \sin nx\, dx$, or (c) $\int \cos mx \cos nx\, dx$, use the corresponding identity:

$$\text{(a)}\quad \sin A \cos B = \tfrac{1}{2}[\sin(A - B) + \sin(A + B)]$$

$$\text{(b)}\quad \sin A \sin B = \tfrac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\text{(c)}\quad \cos A \cos B = \tfrac{1}{2}[\cos(A - B) + \cos(A + B)]$$

EXAMPLE Trigonometric functions with different angles

Find $\int \cos(3x)\ \cos(7x)\ dx.$

SOLUTION

$$I = \frac{1}{8}\sin(4x) + \frac{1}{20}\sin(10x) + C$$

EXAMPLE

You can do it

Find $\int \frac{\tan x \ \sec^2 x}{\cos x}\, dx.$

$$I = \frac{1}{3}\sec^3 x + C$$

Solve exercises of section 8.3.

Start with the suggested problems mentioned in the syllabus.